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Did the universe evolve?†

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Abstract. A new type of explanatory mechanism is proposed to account for the fact that many of the dimensionless numbers which characterize particle physics and cosmology take unnatural values. It is proposed that all final singularities ‘bounce’ or tunnel to initial singularities of new universes at which point the dimensionless parameters of the standard models of particle physics and cosmology undergo *small* random changes. This speculative hypothesis, plus the conventional physics of gravitational collapse, together comprise a mechanism for natural selection, in which those choices of parameters that lead to universes that produce the most black holes during their lifetime are selected for. If our universe is a typical member of the ensemble that results from many generations of such reproducing universes then it follows that the parameters of our present universe are near a local maximum of the number of black holes produced per universe. Thus, modifications of the parameters of particle physics and cosmology from their present values should tend to decrease the number of black holes in the universe. Three possible examples of this mechanism are described. In inflation models I show that, given the hypotheses, there is selective pressure for the very small values of scalar field self-coupling required for inflation. The second concerns the effect on changing the proton–neutron mass difference on the rate of black hole formation in our galaxy. I argue that changing the sign of the mass difference results in a large decrease in the number of black holes produced. Whether raising it results in a decrease or an increase in black hole production is difficult to determine because of the intricate physics of the star formation process. The third mechanism is that in cold, or tepid, big bang models, changes in the proton–neutron mass difference could strongly effect the evolution of the universe as a whole.

1. Introduction

Many, if not most, of the most basic puzzles of particle physics and cosmology involve problems of the *specialness* or *unnaturalness* of the dimensionless parameters that govern fundamental phenomena. These include, on the particle physics side, the gauge hierarchy problem and the strong CP problem, and on the cosmology side, the horizon problem, the flatness problem, and the isotropy problem [1]. In addition, there are problems which involve both particle physics and cosmology such as the problem of the cosmological constant (for a recent discussion see [2]) and the problem of fine tuning of the parameters of inflationary models (for reviews see [3]) in order that they can give an account of the cosmological fine tunings [3–7]. To this list might also be added the problem of the neutron–proton mass difference, as this is presently understood to

† This is a revised version of a preprint initially circulated in February 1990.

arise from the smallness of the light quark masses compared to the QCD scale, and thus may also be thought of as a problem of a large ratio of scales.

In this paper I would like to present a new approach to these problems. This approach is motivated by the observations that not only do so many of the dimensionless parameters in particle physics and cosmology take unnatural values, the values that they do take result in a universe that is much larger, and that contains more structure, than would be the case were these parameters to take more natural values. For example, for natural values of λ , the scalar field self-coupling in inflationary models, the universe would never be larger than Planck scales or grand unified scales, however for λ less than some small critical value the universe can be 60 orders of magnitude larger [3-7]. Similarly, one may note that it is very largely the existence of stars that accounts for the tremendous variety of phenomena that we find in our universe. They are responsible for both synthesizing the higher nuclei and for keeping large regions of the universe far from thermal equilibrium. Stars, in turn, exist due to at least two unnaturally large ratios in the scales that govern elementary particle physics. These are the grand unified gauge hierarchy, which is responsible for the ratio of 10^{40} between the gravitational and the other interactions of nucleons, and the fact that the neutron-proton mass difference is less than nuclear binding energies. Thus, one way to put the question I would like to address in this article is: Why are the laws of physics and the initial conditions of the universe such that stars exist?

One often finds in the literature two opposing views about this kind of problem. The anthropic principle summarizes a collection of observations: the universe we find ourselves in is very special in ways that make it more likely that we, or living things in general, exist. However, it has no explanatory power, given that we exclude explanation by final causes. What is missing is a *mechanism* to explain why it is probable that the (or even a) universe similar to the one we see around us exists.

The opposite attitude is that it is only coincidence that the laws and initial conditions are such as to make the universe hospitable to life. This attitude is an almost necessary consequence of the conjecture that there exists a fundamental theory at the Planck scale that uniquely determines all the coupling constants and masses of fields at all scales. If this is the case then whatever is responsible for the proton-neutron mass difference cannot be causally influenced by the existence or non-existence of a large number of stable nuclei, as this occurs at scales 22 orders of magnitude larger.

The main purpose of this article is to show that there is a middle ground between these two alternatives, i.e. that we can have a scientific and causal explanation of why the universe is such that stars exist. What is needed for such an explanation is two things. The first is a theory in which the fundamental constants in the standard models of physics and cosmology are variable. The second is a dynamical mechanism under which the expressions of these fundamental parameters in relatively low energy physics, such as the existence or non-existence of stars, can effect the probability distribution for those parameters[†]. I will not provide the fundamental theory which allows the fundamental constants to change. Instead, I will show that, given some very general assumptions about how that theory works, there is a mechanism for how the low energy physics could affect the choices of the parameters that requires only that certain hypotheses are made about what happens when a region of the universe collapses to a spacetime singularity.

[†] An alternative possibility is to posit that the parameters are set by a dynamics which extremizes a Lagrangian which depends on a mathematical measure of the structure or variety in the universe [8].

2. A mechanism for natural selection in cosmology

What I would like to propose is a theory of the evolution of the entire universe, including the parameters that govern its basic physics, via a mechanism of natural selection. Now, as in biology, an explanation by means of natural selection of the occurrence of a particular property must have two components: a random component, by which the parameters that determine this property change randomly, but slowly, over many, largely similar, systems and a selective component by which those systems with a particular property are, after some time, more likely to occur. As any physics in which the parameters of the standard model change must be new physics, the random mechanism proposed here is speculative. But the selective mechanism will be based on known physics. In fact the selective mechanism will simply be gravitational collapse. The mechanism is contained in the following two postulates concerning the physics near spacetime singularities.

(i) Each final singularity is followed by an initial singularity, which evolves into a universe which is spatially closed. An alternative hypothesis, which is equivalent as far as its consequences for the subject of this paper, is that instead of an ending in a final singularity, the interior of a black hole tunnels into a new spatially compact universe. We may note that this hypothesis has been advocated as a resolution for the problem of information loss in black hole evaporation [9] and has also been discussed recently in connection with the baby universe scenario [10].

(ii) Let $\mathcal{P}$ be the space of dimensionless parameters which governs the particle physics and matter content of a set of possible universes. A given universe is then associated with a particular point p of this space. Let us then postulate that at each initial singularity (or the tunnelling event) p changes by a small random amount. Thus, there is some probability distribution, $P(p_{\text{old}}, p_{\text{new}})$ which governs the random change in the parameters. One function of a fundamental theory must then be to give us this probability distribution. However, to show that it is plausible that an evolutionary mechanism acted in our universe we only need to assume that $P(p_{\text{old}}, p_{\text{new}})$ exists, and that the average step δp is sufficiently small.

In order to show how these two postulates lead to a mechanism of natural selection I will make one further assumption, which is that we will restrict the parameter space $\mathcal{P}$ to universes that are compact and have positive pressures, so that they always recollapse. Then each universe will have at least one final singularity, and thus at least one progeny.

Now, let us consider two functions which may be defined on $\mathcal{P}$. The first is $T(p)$, which will be defined to be the expected lifetime of a universe with the parameters p . Given my last assumption, this is always finite. The second function is $R(p)$, which is defined to be the expected number of final singularities for a universe with the parameters p . This is equal to one if there is only a final collapse singularity, and is larger if there are black holes. Note that both of these are expectation values defined over an ensemble of universes with parameters p .

It is to be expected that both of these functions are strongly dependent on p . For the reasons we have discussed earlier, it is likely that for most p , $T(p)$ is of the order of t_{Planck} and $R(p)$ is equal to one. However, there will be small regions of $\mathcal{P}$, such as one around the values we find in our present universe, where $T(p) \gg t_{\text{Planck}}$, and $R(p) \gg 1$.

† Note that I am not assuming that $T(p)$ and $R(p)$ are always so correlated. For my purposes I only need assume that there do exist local extremals of $R(p)$ where $T(p)$ is also large.

Now I can explain the mechanism of natural selection which follows from my premises. Let us begin with an initial universe, which is specified by a random point, $p_{\text{initial}} \in \mathcal{P}$. For a typical random p_{initial} , $R(p_{\text{initial}}) = 1$ and the universe produces just one progeny, with a slightly different p . Thus, in the beginning the universes exercise a random walk in $\mathcal{P}$ with diffusion constant $(\delta p)^2/t_{\text{Planck}}$. However, eventually this random walk will wander into a region with $R(p) \gg 1$, after which many universes are created in each generation. These universes may be described by a distribution of points, $\rho(p)$, in $\mathcal{P}$, whose number continues to increase in number with each generation. More specifically, we will be interested in $\rho(p, N)$, which is the distribution of universes which have been created in the N th generation, or earlier, and in the normalized limit

$$\Delta(p) = \lim_{N \rightarrow \infty} \frac{\rho(p, N)}{\int dp \rho(p, N)}. \quad (1)$$

Now, we may observe that the following are likely consequences of the assumptions we have made so far: (i) the limit $N \rightarrow \infty$ in the definition of $\Delta(p)$ converges, so that this function is well defined; and (ii) $\Delta(p)$ is peaked around local maxima of $R(p)$ the expected number of black holes.

If the ideas proposed in this paper prove fruitful it will be important to show that (i) and (ii) do, indeed, follow in an appropriate set of models. For the moment it is sufficient to assume them and investigate their consequences. To do this, and indeed in order to connect any results about these multi-universe distributions to observations, we have to make one final assumption, which is that we are in a typical universe in the ensemble defined by $\Delta(p)$.

We have a mechanism which will allow us to make non-trivial predictions because it follows from the foregoing that our universe should have parameters p that are near a local maximum of $R(p)$, the expected number of black holes. Thus modifications in any of the parameters should generally result in a decrease of $R(p)$. Thus, if we can calculate $R(p)$, just for p close to the parameters of our universe, we have arrived at a prediction that we may expect could be verified or refuted given known physics: *the parameters of particle physics are such that most changes in their values should lead to decreases in the expected number of black holes in the universe.*

Note that to verify this prediction we do not need to be able to calculate $R(p)$ for all p , we only need to be able to calculate it for p near to the parameters of our own universe. It then becomes interesting to know to what extent $R(p)$ is computable in the neighbourhood of the p of our universe given known physics and astrophysics, I will comment on this in section 5. Note also, that because we assume that we are in a typical member of the ensemble, and hence in a universe whose p is close to a *local* extremum, to extract predictions from the theory we need know nothing about the initial point p_{initial} . Indeed, we need to know nothing about the actual $\Delta(p)$ except that it is peaked around local extrema.

Finally, suppose we knew enough about $R(p)$ that we could in fact calculate the distribution $\Delta(p)$ on all of $\mathcal{P}$. Could we use this information to go further and make global predictions about the constants of the standard model? This depends on the nature of the distribution $\Delta(p)$. It may or may not be the case that $\Delta(p)$ has two additional properties: (iii) the final $\Delta(p)$ is independent of initial point p_{initial} ; and (iv) $\Delta(p)$ is strongly peaked around one global extremum. If we can calculate $R(p)$ everywhere, and if it is then the case that the resulting $\Delta(p)$ has these properties, then we could, in principle, predict the values of the parameters p . Having said this I would like to stress that, while to compute $R(p)$ everywhere is a difficult problem, it

is only necessary to compute $R(p)$ for p near the parameters of our present universe to test, or falsify, this theory.

Before going on to discuss how this proposed mechanism might be applied to specific problems in particle physics and cosmology I would like to make four general comments.

First, it is important to stress that here, as in biology, natural selection is not a tautology. It is a tautology that that which is more stable survives longer. However, to have a mechanism that has a probability close to one to actually generate structure in some given finite time, several conditions need to be satisfied. The rate of mutations must be fast enough that many mutations can take place within the time and space over which the evolution may be known to have taken place. However, the change in the expression of the parameters (or genes) due to each random change must be small. This is because if the subspace of the parameter space $\mathcal{P}$, which leads to universes with many progeny is very small, and if $R(p)$, the number of progeny, is, within this subspace, a sensitive function of p , small mutations (random changes in p) are much more likely to be successful, i.e. to lead to increases in $R(p)$ than large mutations.

If these conditions are not met, for example, if the random changes are too large or too infrequent, then although natural selection will operate it is improbable that any structure significantly far from thermal equilibrium will emerge.

Second, I have argued that the hypothesis of natural selection can have predictive power even in the absence of a fundamental theory of the mechanisms behind the random element. However, it is possible that, just as occurred in biology, at a later date a detailed mechanism for this random component might emerge. This is not as unlikely as it sounds, indeed the possibility that the parameters of the standard model are determined dynamically is prevalent in much recent work on string theory (see for example [11]), inflationary cosmology [13] and wormhole physics [14]†.

For example, suppose that the laws of nature are governed by some perturbative string theory, perhaps because these are the only solutions to the problem of combining consistently in a perturbative framework quantum theory and gravity. However, there may be no physical principle which can select any of these preferentially over any other. Certainly there is no principle of minimizing energy, as in a closed universe there is no physically meaningful definition of energy. Thus all perturbative string theories may be equally favoured, in which case it perhaps makes sense to say that one is chosen, randomly, at the beginning of the universe.

Alternatively, there has been much interest in recent years in the wormhole hypothesis, within which the values of the dimensionless parameters in particle physics become dynamical variables whose values involve dynamics at Planck scales [12]. In such a scenario it would not, perhaps, be unlikely, that when spacetime curvatures become of the order of the Planck scale these constants could be subject to small changes.

The third, and crucial, point is that the parameters which govern a phenomenon at

† Linde [13] makes a proposal which in some respects relates to that discussed in this article; the crucial difference is in his use of the anthropic principle in place of the present argument, which is based on a proposal of an explicit mechanism for selective pressure. For the anthropic principle to work there must exist a universe like ours somewhere in an ensemble generated randomly by the initial distribution of the scalar field in the inflationary model, whereas for the present argument to work something much stricter is required, which is that our universe be a typical member of an ensemble whose distribution is determined by the selective rates of reproduction of universes with different parameters.

some scale should be determined by selective mechanisms acting at that scale. This is just a consequence of naturalness. This assumption, together with the assumption that the changes from generation to generation must be small lead us to the conclusion that there should have been a succession of stages of evolution of the parameters, such that the parameters that fix phenomena at successively lower energy scales are determined at successively later stages.

Thus, during the i th stage, the ps will begin in a region $\mathcal{R}$ of $\mathcal{P}$ associated with universes that expand until some scale of lengths l_i and densities ρ_i . During this stage the universes will explore which phenomena may take place at these scales as they explore $\mathcal{R}$ and nearby regions of $\mathcal{P}$. If in these regions there are parameters for which a universe will produce more black holes than other universes, these parameters will be selected for. This can occur either through a universe with those parameters living much longer than l_i/c or by adjustment of the particle physics parameters. However, in the former case the progeny will then expand to a new scale $l_{i+1} > l_i$, and a new stage of the process will begin in which phenomena on the new scale l_{i+1} will be explored.

If current notions of grand unified theories are correct, these stages might include some associated with (i) the dimensionality of space, (ii) choice of the gauge group, (iii) choice of the parameters that lead to inflation, (iv) choice of parameters that lead to baryosynthesis, (v) choice of parameters that lead to electroweak symmetry breaking and (vi) choice of parameters that lead to nucleosynthesis.

Finally, on general grounds, it is plausible that there exist local maxima of $R(p)$ associated with large universes with a great deal of structure, of the sort that we see around us. Let us consider p which have large $T(p)$ and in addition have the property that there is a large gauge hierarchy, which is to say a large ratio between the Planck mass and the mass of the lightest stable particle. (In the next section I discuss how the proposed mechanism might lead to such conditions.) In such a universe, black holes can only be produced by gravitational collapse, and this requires the existence of gravitationally bound systems. Thus, for universes with large $T(p)$ and large gauge hierarchies, $R(p)$ should be roughly proportional to the number of gravitationally bound systems. Thus, to the extent that the number of gravitationally bound systems in a universe can be taken as a measure of the 'structure' of that universe, it may be that this mechanism also selects for universes with more structure.

In the following sections I discuss three examples of how the general principles stated in this section might work out. I want to stress that what follows are only very rough sketches of how the postulates introduced in this section might be applied. Whether these ideas are really viable will depend on whether detailed and realistic models along these lines can be developed which lead to predictions which are falsifiable by observations or experiments.

3. Natural selection for inflation?

The inflationary hypothesis is attractive from several points of view, in that it in principle eliminates several problems of fine tuning in cosmology, such as the horizon problem, the flatness problem and the monopole problem [3]. There are several different kinds of inflationary models here, and it is not my purpose here to review or assess them. For the present purposes it is interesting to note that in some kinds of inflationary models the effect can only be achieved if certain dimensionless couplings

are finely tuned to very small values. From the point of view of inflationary theorists, this has always been seen to be a liability, and several of the newer inflationary models, such as chaotic inflation, were invented to eliminate this fine tuning problem [3]. However, it is exactly these older inflationary models with fine tuning problems that I will be interested in here, because they offer an opportunity for a mechanism of the kind we have just outlined to operate.

There are two kinds of fine tuning problems that one finds in these inflationary models. The first of these is that for inflation to occur at all in these models the self-coupling of the scalar field whose vacuum expectation value drives inflation must take an unnatural value of $\lambda < \lambda_{\text{critical}}$ in order to avoid the objection of Mazenko, Unruh and Wald [4, 5]. The exact value of $\lambda_{\text{critical}}$ depends on other parameters of the scalar field potential [7], but for typical values it is very small. In addition, the scale of density fluctuations, $\delta\rho/\rho$, predicted by inflationary models also show a dependence on $\lambda < \lambda_{\text{critical}}$. In realistic models of inflation this means that if $\delta\rho/\rho$ is to satisfy the constraint which follows from the observed level of isotropy of the microwave background λ must be less than 10^{-12} [3, 5].

The second problem is that a mechanism that explains the value of λ must explain not only why it is so small. It must also explain why it is not still smaller. A possible explanation for this may lie in the dependence of $\delta\rho/\rho$ on λ just mentioned [3, 5, 14]. If $\delta\rho/\rho$ is too small then the fluctuations may never become large enough for a significant number of black holes to condense in the lifetime of the universe. Thus, the mechanism of natural selection that extremized $R(\lambda)$ may select for a λ that is a compromise between two competing tendencies. If λ is too large then inflation does not occur, or does not last long enough for many black holes to evolve from the fluctuation spectrum $\delta\rho/\rho$. But if λ is too small then the fluctuations may not be large enough to seed black holes. It is interesting to speculate whether in the context of a particular model the mechanism for natural selection that extremizes R might predict a value for $\delta\rho/\rho$ consistent with present limits on the black body radiation [15].

There is a third fine tuning problem that is more general than inflationary models, and afflicts all attempts to join particle physics with general relativity and cosmology. This is the problem of the cosmological constant. In the context of inflationary models this problem becomes the question of why the vacuum expectation value of the potential energy after the last of, perhaps, several stages of symmetry breaking, is exactly zero? Unfortunately, the problem of the cosmological constant is very difficult, as it is fine tuned to energy scales which are much smaller than those associated with the other fine tunings. I have not been able to invent a selective mechanism that might explain the smallness of the cosmological constant. However, I would like to show that it is natural to solve the two fine tuning problems which appear in inflationary models through the selective mechanism described in the previous section.

To see this we construct a simple model of a universe with one scalar field. We assume that the parameter space $\mathcal{P}$ is one-dimensional, and is parametrized by the self-coupling of a scalar field, λ . We will assume that we are working with a model subject to the analysis of Mazenko *et al* [4] so that inflation is not possible unless λ is very small, say $\lambda < 10^{-6}$. We will also assume that when inflation occurs a universe blows up to a size,

$$L(\lambda) = F(\lambda)L_{\text{Planck}} \quad (2)$$

before recollapsing. The simplest choice for $F(\lambda)$ is that it is equal to 1 for $\lambda > \lambda_{\text{critical}}$ and equal to $\exp(B)$ for $\lambda < \lambda_{\text{critical}}$ where B can for a first approximation be taken

to be λ independent.

We will assume that if a universe becomes big enough compared to L_{Planck} gravitational clumping occurs during its lifetime, so that it collapses inhomogeneously to many final singularities. Following the two rules we posited previously, each of these final singularities becomes the initial singularity of a new universe. The number of new universes which are thus formed should be proportional to the volume of the parent universe, when the new universe grows to a scale large enough, and lives long enough, that gravitational clumping can occur and carry through to collapse. Thus we will assume that the number of new universes is

$$R(\lambda) = 1 \quad \text{if } L(\lambda) < L_{\text{critical}} \quad (3)$$

and

$$R(\lambda) = K[L(\lambda)]G[\delta\rho/\rho(\lambda)] \quad (4)$$

for $L(\lambda) > L_{\text{critical}}$. Here $K[L(\lambda)]$ is expected to be a monotonically increasing function of the volume L^3 which expresses the fact that, everything else being equal, universes that are larger and live longer will have more black holes. For example, we might take in our one parameter model

$$K[L(\lambda)] = C \left[\frac{(L(\lambda) - L_{\text{critical}})}{L_{\text{critical}}} \right]^3 \quad (5)$$

although a more sophisticated model might also involve the temperature of the matter through L/l_{Jeans} , with l_{Jeans} the Jeans length.

The factor $G[\delta\rho/\rho(\lambda)]$ parametrizes the fact that universe which have a larger $\delta\rho/\rho$ will also have more black holes, as $\delta\rho/\rho$ increases with increasing λ this will be a monotonically increasing function of λ .

We then see that in this simple class of models $R(\lambda)$ will have a maximum at $\lambda_{\text{critical}}$. In a more general model there is more than one parameter in the scalar potential, and the expansion factor F for $\lambda < \lambda_{\text{critical}}$ will have some dependence on these parameters. The maximum of R will, however, still be expected to come from a competition between the two effects mentioned previously.

To complete the definition of the model, we need to specify the probability distribution for the random change in λ at each initial singularity, the parameters L_{critical} and C and the form of the function K . The model seems more sensible if we assume that at each new singularity it is $1/\lambda$ that takes a random step, of $\delta(1/\lambda) \approx 1$. We must assume also that there is a reflecting barrier at $1/\lambda = 0$ so that λ does not become negative. Given this and the other parameters we have a model we can run by numerical simulation. However, without specifying the other parameters we can see what will happen generally when the model is run. It is natural to begin the first universe with an initial $\lambda_i \approx 1$ so that no inflation occurs. The first universe lives a Planck time, and then gives rise to a single new universe, with a new λ^{-1} . Each new λ will involve a random step of $\delta(1/\lambda) \approx \pm 1$. Thus, $1/\lambda$ undergoes a random walk with a diffusion constant $D = 1/t_{\text{Planck}}^2$. Until $\lambda^{-1} > \lambda_{\text{critical}}^{-1}$ each universe lives for a Planck time and then reproduces itself. However, after $\lambda_{\text{critical}}^{-2}$ Planck times it will become probable for $\lambda < \lambda_{\text{critical}}$ so inflation will occur. The first inflating universe thus has $\lambda \approx \lambda_{\text{critical}}$, and it has many progeny. There will now be a distribution of λ in the progeny, which is originally peaked around $\lambda_{\text{critical}}$. If we continue for an arbitrary

number of further generations the distribution stays peaked around $\lambda_{\text{critical}}$ because $R(\lambda)$ is peaked around $\lambda_{\text{critical}}$. Thus, the model predicts that no matter what generation it is, if we live in a universe that inflated we live in a universe with $\lambda \approx \lambda_{\text{critical}}$.

Thus, this simple model makes a prediction, which is that we will observe fluctuations in the universe of the order of $\delta\rho/\rho(\lambda_{\text{critical}})$. This cannot yet be taken seriously; however it suggests that it might be possible to elaborate this model to the point where a serious prediction could be made for $\delta\rho/\rho$. To do this we need three things, a realistic inflationary model, reliable estimates of $\lambda_{\text{critical}}$ in these models and realistic estimates, based on known astrophysics, for the functions K and G . We may note that numerical simulations to determine $\lambda_{\text{critical}}$ as a function of various parameters in inflationary models have been done [7]†.

4. Natural selection for a small proton-neutron mass difference?

The current understanding of the proton-neutron mass difference is that it cannot be explained by electromagnetic effects, and must therefore reflect a mass difference between the up and down quarks. Since $(m_{\text{up}} - m_{\text{down}})/m_{\text{down}}$ is of order one, the problem of the proton-neutron mass difference is first to explain why $(m_n - m_p)/m_n \approx (m_{\text{down}} - m_{\text{up}})/m_n \approx m_{\text{down}}/m_n$ is a small number. The second thing to explain is the sign, why $m_n > m_p$. I would like now to investigate whether these two facts could have an explanation in terms of a mechanism of natural selection as outlined previously.

According to the general ideas described previously, the parameters of the standard models of particle physics and cosmology are to be determined by selective effects acting at the natural scales associated with the physics governed by those parameters, in decreasing order of those scales. Thus, if the proton-neutron mass difference is to have an explanation via the mechanism of natural selection outlined earlier two assumptions can be made: (i) the mechanism must involve physical processes at a scale of a few MeV; and (ii) we can assume that when this process is important all the parameters associated with larger energy scales have been fixed by previous stages of the evolution.

Now, a basic observation is that the existence or non-existence of a large number of stable nuclei is sensitively tied to the proton-neutron mass difference. We may recall that the binding energy per nuclei in helium is 7 MeV, thus, were the $m_n - m_p > 14$ MeV helium would not bind. Furthermore, the binding energy per nuclei increases by only one MeV as we go up from helium through all the heavier nuclei. Thus, it seems very likely that if $m_n - m_p$ were larger than, say, 18 MeV there would be no stable composite nuclei. Now the real $m_n - m_p$ is 1.29 MeV, which is a factor of 14 less than that. However, we should recall also that deuterium plays a crucial role in the processes by which the heavy elements were formed, in both the primordial synthesis of helium and in stars, and deuterium, with a binding energy of 2.23 MeV would not exist were $m_n - m_p$ twice as big.

† One final note about inflation. According to this general framework outlined earlier we may look for a mechanism of selective pressure for each tunable parameter in a given model of particle physics. Inflation is clearly the optimum selective mechanism for tuning parameters associated with spontaneous symmetry breaking; this suggests that inflation could have occurred in several stages, each associated with different symmetry breaking scales [21]. If there is selective pressure to make the ratios of the symmetry breaking scales large, the evolutionary mechanism proposed here may be able to explain the gauge hierarchy problem.

Now, if there were no deuterium (or, certainly if there were stable nuclei besides hydrogen) then there would be no energy production in stars and no era of nucleosynthesis in the early universe. So the natural question to ask is whether we can find a mechanism having to do with one or the other of these two processes that, acting in the context of the two postulates of section 2, would function as a selective mechanism for the proton-neutron mass difference to be small enough for deuterium to exist.

We begin with the stars.

5. Natural selection for the existence of stars?

To investigate whether there might be a mechanism for natural selection realizing the scenario discussed above which would involve the existence of stars we are making use of a hypothesis, which is that over the whole lifetime of our universe the primary mechanism of black hole production is stellar collapse. Making this assumption, we then want to compute $R(\Delta m, p)$, where R is the number of black holes in the universe, $\Delta m = m_n - m_p$ and p represents all of the other parameters of the standard models of particle physics and cosmology. We will assume that all of these other parameters can be held fixed as Δm is varied. Further, in order not to rely on global assumptions about the universe, we would like to calculate not $R(\Delta m, p)$, but the number of black holes produced during the lifetime of the universe per 10^{12} solar masses, the mass of a typical galaxy, which will be denoted $G(\Delta m, p)$.

Our hypothesis, which I would like to now briefly discuss, is that with the other parameters fixed, $G(\Delta m, p)$ has a local maximum at the observed $\Delta m = 1.29$ MeV.

To get started, we will assume that varying Δm has no effect on the formation of galaxies so that we begin with 10^{12} solar masses of gas and dust, condensed into a rotating distribution about the size of our galaxy. Having made all of these assumptions, it is then clear that the calculation of $G(\Delta m, p)$ is an exercise in stellar physics.

I would like to present a very simple model of how $G(\Delta m, p)$ might be computed. This model makes several assumptions that I will argue shortly are wrong. However, by going through the exercise of constructing and analysing this model, we will see exactly what we will need to know to make a reliable calculation of $G(\Delta m, p)$. The first input we need for this model is called the stellar birth rate function, and is denoted $D(\mu, \Delta m, p, t)$. It is defined so that $D(\mu, \Delta m, p, t)d\mu$ is proportional to the number of stars in a galaxy that are born with masses between μ and $\mu + d\mu$ at time t . We have indicated its dependence on Δm and the other parameters of particle physics, which we would like to calculate.

Another function of interest is the present day mass function, which I will denote $P(\mu, \Delta m, p, t)$. It is defined so that $P(\mu, \Delta m, p, t)d\mu$ is equal to the number of stars with masses between μ and $\mu + d\mu$ in a galaxy of 10^{12} solar masses at the time t . Given information about the lifetimes and evolutions of stars, the two functions can be related to each other, as is described in [16, 19].

A great deal of work has gone into the measurement of the present day mass function, in our galaxy, at the present time. There are also observations which give us information about this function for a number of other galaxies. A good review of the state of the art concerning initial mass functions is in [19]. On the theoretical side, one of the basic goals of the theories of star formation and galactic evolution is to give a calculation of the stellar birth rate function, from first principles. This has not yet been accomplished. Thus, for the present we must take for $D(\mu, \Delta m, p, t)$ values

which are inferred from observations of the present day mass function. This only gives us the values of the function for the actual parameters of the universe. This leaves us with two problems if we are to try to test our hypothesis that $G(\Delta m, p)$ has a local maximum at the present Δm . First, we must be able to deduce something about the way $D(\mu, \Delta m, p, t)$ changes as we change Δm . Second, we must be able to deduce something about the variation in time of $D(\mu, \Delta m, p, t)$.

This last question has been the subject of some work by astrophysics. There have been proposals that certain aspects of the observational data can be explained by attributing certain time dependences to the initial mass function [20]. However, there does not appear to be yet a conclusive argument about this, and it seems that nothing in the experimental data is strongly contradicted if we make the assumption that $D(\mu, \Delta m, p, t)$ is constant in time. I will make this assumption in order to proceed with the construction of a model, but I will come back in the end to criticize it.

The next input that is needed is a function $\rho(\mu)$, which is defined to be the proportion of stars with mass μ which will collapse to black holes. Unfortunately, not very much is known about this function, either theoretically or observationally. It is clear that for μ below the Chandrasekhar mass $\rho(\mu)$ must vanish, and it is also considered rather likely that there is a cutoff mass μ_{bh} considerably higher than the Chandrasekhar mass below which ρ vanishes. This is because a massive star is expected to lose a large fraction of its mass to mass loss during its evolution as well as to supernova before collapsing to a final neutron star or black hole.

Unfortunately, the theory of supernovas is not sufficiently developed to give a theoretical determination of $\rho(\mu)$ beyond these schematic comments. Since black holes are hard to see, there is also little that can be deduced from the observational side. It seems consistent with present ignorance to assume that there is some mass μ_{cutoff} such that all stars with larger masses become black holes, while all stars with lower masses do not. Shapiro and Teukolsky [16] take μ_{cutoff} to be 10 solar masses. We may note that this leads to an estimate that there is approximately one black hole per 10^3 cubic ps in our galaxy [16], so that our universe is indeed a prolific progenitor of black holes.

Let us then construct a model in which a first generation of stars form out of an initial contracting cloud of gas and dust as the galaxy forms, and succeeding generations are formed from the matter which is ejected or lost by previous generations†. The number of black holes which form in the first generation is then proportional to,

$$b(\Delta M, p) = \int d\mu D(\mu, \Delta m, p) \rho(\mu). \quad (6)$$

This is not yet the number of black holes produced during the lifetime of our universe because massive stars return a large portion of their mass to the interstellar medium by supernova and mass loss, and this mass then condenses into further stars. We then need another input to our model, which is the proportion of mass that stars of mass μ return to the interstellar medium. We will denote this by $\delta(\mu)$. Now, if we

† We may note that in this model a generation corresponds to the number of times that material has been recycled in stars, and does not require that the first generation forms all at once, the second generation at some fixed time after that and so on. This is because we are not interested in calculating the present day mass function at a given time, but only the number of black holes that are formed at the end of the whole process.

normalize $D(\mu)$ so that

$$M_{\text{total}} = \int d\mu \mu D(\mu) \quad (7)$$

is the total mass of gas and dust that goes into the first generation (which means the total mass that goes into stars at any time during the lifetime of the galaxy), then

$$\Delta(\Delta m, p) = \frac{\int d\mu \mu D(\mu) \delta(\mu)}{M_{\text{total}}} \quad (8)$$

is the proportion of that mass which will be returned by the first generation. Let us now assume that this returned mass will be recycled entirely into stars, with a mass distribution given by the initial mass distribution $D(\mu)$ (which we have assumed to be time independent). Then the total number of black holes produced by all generations of stars in our galaxy is given by

$$G(\Delta m, p) = b(\Delta m, p)(1 + \Delta + \Delta^2 + \dots) = \frac{b(\Delta m, p)}{1 - \Delta(\Delta m, p)}. \quad (9)$$

We can use this simple model to discuss whether the number of black holes in the galaxy goes up or down when we change Δm . First, let us consider what happens if we change the sign, so that the proton is heavier than the neutron.

I would like to argue that in this case the number of black holes decreases dramatically. The reason is that for star formation to occur at all, there must be a mechanism to allow the initial cloud of gas and dust to lose energy so that it can condense. For clouds of hydrogen and helium gas the primary mechanism for such energy loss is radiative transfer. Were the neutron, rather than the proton, to be the lightest baryon, the galaxy would consist initially of mostly a gas of neutrons, rather than hydrogen. As the coupling of neutrons to the electromagnetic field is weaker by a factor of α , the fine structure constant, than the coupling of hydrogen, the process of radiative transfer, and hence of condensation would be that much slower.

Thus, we may conclude that $G(\Delta m < 0) \ll G(\Delta m > 0)$. Under the hypotheses outlined in section 2, this can be taken as a possible explanation of the fact that the neutron is heavier than the proton. We may also note that this conclusion is independent of the details of our oversimplified model, it uses only the fact that radiative transfer is the dominant mechanism of energy loss in the condensation of clouds of hydrogen gas leading to star formation.

Let us now consider what happens when we increase Δm , keeping the sign positive. The simplest thing to do is to compare the present $G(\Delta m = 1.29 \text{ MeV})$ with the case $G(\Delta m = 18 \text{ MeV})$, in which case there would be no stable nuclei besides hydrogen, and hence no nuclear fusion. In this case there is no stellar evolution and no supernovas. Although this does not rule out processes of mass loss arising from pure hydrodynamics, we will in this simple model neglect those processes, so that we assume that any condensing cloud of hydrogen gas more massive than the Chandrasekhar mass collapses to a black hole. Thus, we have $\rho(\mu, \Delta m = 18 \text{ MeV}) = \theta(\mu - \mu_{\text{Chandra}})$. Since there is no mechanism for stars to return significant amounts of matter to the interstellar medium, we have simply, $\delta(\mu, \Delta m = 18 \text{ MeV}) = \Delta(\Delta m = 18 \text{ MeV}) = 0$.

We will model the throwback function for our present universe by the simple assumption that every star more massive than some value μ_{SN} supernovas, and that when it does so it returns all but its Chandrasekhar mass to the interstellar medium.

It is my understanding that this assumption is neither strongly supported nor strongly contradicted by current astrophysical knowledge. We thus have

$$\delta(\mu) = \theta(\mu - \mu_{\text{SN}}) \frac{(\mu - \mu_{\text{Chandra}})}{\mu}. \quad (10)$$

Finally, recall that for our present universe we are making the simple assumption that $\rho(\mu, \Delta m = 1.29 \text{ MeV}) = \theta(\mu - \mu_{\text{cutoff}})$.

In order to proceed to evaluate equation (9) we need to make some assumption about the dependence of the stellar birth rate function on Δm . We will first consider the simplest assumption, which is that $D(\Delta m, p)$ is actually independent of Δm , so that in our hypothetical world with $\Delta m = 18 \text{ MeV}$ it has the same form as is observed in our present universe. I will postpone a discussion of the plausibility of this assumption until after we have seen its consequences. For a first calculation we may use the Salpeter [17] form of the function, which is

$$D(\mu) = \left(\frac{\mu}{\mu_0} \right)^{-\beta} \quad (11)$$

where β is about 2.35 [16]. It is then straightforward to calculate

$$Z = \frac{G(\Delta m = 18 \text{ MeV})}{G(\Delta m = 1.29 \text{ MeV})}. \quad (12)$$

If $N_{\text{bhp}} = \int_{\mu_{\text{cutoff}}}^{\infty} D(\mu) d\mu$ is the number of black hole progenitors (in our universe) in the present day mass function we then find that

$$Z = \left(\frac{\mu_{\text{cutoff}}}{\mu_{\text{Chandra}}} \right)^{\beta-1} \left[1 - \left(\frac{\mu_{\text{SN}}}{M_{\text{total}}} \right) N_{\text{bhp}} \left(\frac{\beta-1}{\beta-2} \right) \left(1 - \frac{\mu_{\text{Chandra}}}{\mu_{\text{SN}}} \left(\frac{\beta-1}{\beta-2} \right) \right) \right]. \quad (13)$$

If we recall that $\mu_{\text{cutoff}} > \mu_{\text{SN}} > \mu_{\text{Chandra}}$ we see, first of all, that the first factor is larger than one; if we take $\mu_{\text{cutoff}} = 10$ solar masses we have for this factor about 10. Thus, Z will be greater than unity unless the quantity in brackets is less than about 1/10. But this is exceedingly unlikely, as for any reasonable values $(\mu_{\text{SN}}/M_{\text{total}})N_{\text{bhp}} \ll 1$ due to the rapidly falling character of $D(\mu)$. Thus, we conclude that the recycling of matter in the present universe does not make up for the fact that many more 'stars' collapse immediately to black holes in the hypothetical universe, and our hypothesis is contradicted, given the assumptions that went into our simple model.

Let us now consider the assumptions that went into this model. Besides several simplifying assumptions, which in any case are within the bounds of present astrophysical knowledge, the main physical assumption that went into this calculation is that the stellar birth rate function is independent of the neutron-proton mass difference. Is this a reasonable assumption? I would like to argue that it is not.

In order to investigate this question we have to look into the physics of star formation. The assumption in question would be reasonable if that physics involved only hydrodynamics and radiative transfer, that is if it involved only collapsing clouds of hydrogen gas and electrodynamics. However, present understanding of the star formation process is that this is not the case. It is now widely believed that chemical reactions, taking place on the surfaces of the carbon dust grains, play a major role in cooling the clouds in which new star formation is seen to be taking place (see, for

example, [18]). From this we may conclude that the star formation process, which determines $D(\Delta m, \mu)$, depends importantly on the presence of carbon and other organic elements, and hence would be very much altered if Δm were increased. This is not to say that star formation would not take place were there no higher elements, indeed such processes must have taken place as all the higher elements are believed to have been produced in stars. But however that initial generation of star formation occurred, it was not by means of the process of star formation that we see at present. The important conclusion is that there is no reason for the assumption that $D(\Delta m, \mu)$ is independent either of Δm or of time.

Is there any way to determine what initial mass distribution would result from a gas of pure hydrogen and helium? Unfortunately, this cannot be determined by observation, for in any population, such as the halo stars, which may be the result of an earlier mechanism of star formation, all the massive stars are already invisible remnants, and we thus have no information about the distribution function in the region of interest for this calculation. The only way to determine the dependence of the mass distribution function on Δm must be through a much improved theoretical understanding of actual and possible star formation processes.

I would like to close this section with two morals. The first is that to calculate the dependence of $R(p)$ on the parameters of particle physics in the neighbourhood of the present values of the parameters will require a level of precision of our understanding of basic astrophysical processes that has not yet been achieved. However, as astrophysics is at present a rapidly developing subject, it may be hoped that calculations of this kind may become possible in the not too distant future. One of the main themes of this paper is that, under the kind of hypothesis made above, these kinds of calculations, which after all rely only on physics at experimentally accessible scales, could play a crucial role in our understanding of how the parameters in the standard models of particle physics are fixed.

The second moral is that the mechanisms of star formation and galactic evolution are much more complicated than might have been expected. Indeed, the basic problems in the physics of galaxies, such as accounting for the different types of galaxies, or understanding the existence and stability of the spiral arms, are reminiscent of the kinds of problems of self-organization and stability that one studies in diverse fields in which non-equilibrium stable structures are of interest. There seems to be a kind of ecology in the physics of the spiral galaxies by means of which the structures responsible for star formation—the spiral arms and the associated clouds of dust and gas—are maintained for time scales much longer than the relevant dynamical time scales. Indeed, just the facts that star formation is still taking place in spiral galaxies, and that there is still a considerable amount of matter in interstellar clouds of gas and dust at a time much later than the relevant dynamical time scales, suggests that there must be processes of self-organization and self-regularization governing star formation and galactic evolution in spiral galaxies. These must involve self-organizing cycles of materials and energy of the kind that one sees in diverse non-equilibrium systems as well as in biological systems†.

† While these problems take us outside the scope of the present paper, we may observe that an 'ecological' approach to the evolution of galaxies is consistent with the basic theme of this paper.

6. Natural selection during the era of nucleosynthesis

Let us now turn to the question of whether a selective mechanism could have acted during the era of primordial nucleosynthesis. The question we must then ask is, what is the consequence of the fermion mass matrix for the overall evolution of the universe at nucleosynthesis times? The answer depends on whether the universe at the era of primordial nucleosynthesis is radiation or matter dominated. In the standard radiation dominated case, the energy-momentum tensor is dominated by thermal radiation, and the physics of the baryons will have no effect on the evolution of the universe.

However, cosmological models which are matter dominated for the whole evolution of the universe have been proposed, and their consequences have been studied in detail [22]. In these proposals the cosmic black body radiation is produced during a period shortly after decoupling, either in a generation of early 'population III' stars which form and burn out quickly, before galaxy formation, or as a result of radiation by accretion discs of black holes which form at that time.

One advantage of these models is that the problem of accounting for the early growth of inhomogeneities engendered by recent observations of large and early structures is removed†. The problem of nucleosynthesis in these 'tepid' big bang models has also been studied in detail [23]. It was found that there are scenarios in which helium production occurs at the levels of the hot big bang models. Deuterium production at what are considered acceptable levels is, however, more difficult to account for in these models. While this problem does not rule out such models, because of various uncertainties in estimates for production of deuterium after decoupling [24], it is thought to be a major difficulty which needs to be settled if these models are to be taken seriously.

Having settled that there are cold big bang models which are not completely ruled out at the present time we can investigate whether the presence or absence of an era of nucleosynthesis can have a dramatic effect on the evolution of the universe in such a model. To investigate this question we may consider a matter dominated homogeneous cosmology. For the present purposes it will be useful to recall that, as is well known, the equations of this model are isomorphic to the equations that describe a homogeneous sphere of self-gravitating matter in Newtonian physics. Thus, let us consider a homogeneous sphere of radius R and mass M expanding radially. The total of the gravitational potential and kinetic energy of the sphere is given by,

$$E = \frac{M}{2} \dot{R}^2 - \frac{3}{10} \frac{GM^2}{R} \quad (14)$$

or, in FRW form,

$$\frac{\dot{R}^2}{R^2} = \frac{3}{10} \frac{GM^2}{R^3} + \frac{2E}{M R^2}. \quad (15)$$

We see that this is the same as the FRW universe with the time coordinate scaled unconventionally so that $k = -2E/3M$, instead of ± 1 or 0. For $E < 0$ the universe

† The models are also consistent with the limits on anisotropy of the black body radiation, because the radiation produced by the early stars or black holes is then thermalized by the matter which still has a short optical mean free path at this time. Finally, these models provide, in the relics of the early stars or black holes, natural candidates for the dark matter.

expands to a maximal radius R_m and recollapses to a singularity after a total time T ; these are given by

$$R_m = \frac{3}{10} \frac{GM^2}{-E} \quad (16)$$

$$T = \frac{3\pi}{10} \frac{GM^{5/2}}{(-E)^{3/2}} \quad (17)$$

Now, the E in these equations is only the sum of the kinetic and gravitational potential energy. Thus, if at some time t_0 nuclear reactions take place which transfer an amount of energy $\Delta\epsilon$ per baryon from the mass energy of the particles to kinetic energy, the total E will change by $N\Delta\epsilon$. What we want to know is the value of $\epsilon = E/N$ such that our universe, with 10^{80} baryons, can have a lifetime exceeding 10^{17} s. Expressing T in more natural units we have

$$\epsilon = \left(\frac{3\pi}{10}\right)^{2/3} \left(\frac{\hbar N}{T}\right)^{2/3} \left(\frac{m_{\text{proton}}}{M_{\text{Planck}}}\right)^{4/3} (m_{\text{proton}} c^2)^{1/3} \quad (18)$$

Note that the dependence of this quantity on N arises because the total energy (7) is a sum of a linear and quadratic piece in N . Putting in $N = 10^{80}$ and $T = \pi \times 10^{18}$ s we find that ϵ is of the order of 100 MeV. Since the lifetime of our universe could be much larger than the current Hubble time, this means that it is within the realm of possibility that in a matter dominated cosmological model a transfer of the order of 1 MeV per baryon from mass energy to kinetic energy could have an effect on the evolution of a universe similar to ours. This suggests that it is not impossible that a mechanism of natural selection, such as the one outlined previously, could have selected for a proton-neutron mass difference which would lead to an era of nucleosynthesis in which this order of magnitude of energy per baryon was transferred from mass energy to kinetic energy†.

Of course, this model is very crude, for a more detailed model we need to study the dynamics of an inhomogeneous cosmological model, as pressure can only act to divert particles from geodesic motion when there are inhomogeneities. The only conclusion that can be drawn from this simple calculation is that it is not impossible that in a cold big bang model the gross evolution of the universe could have been significantly altered by the presence or absence of an era of nucleosynthesis.

Finally, it must be emphasized that the statement that no nuclear bound states would exist were $|m_{\text{down}} - m_{\text{up}}|$ larger than some value is a hypothesis which needs to be supported by detailed calculations of the effect on the two nucleon potential of modifying the light quark masses.

7. Are the generations fossils of earlier universes?

I would like to end with a speculation about how the quark mass matrix could have changed at each initial singularity. Suppose that at each new universe one generation of fermions is added, but all of the old ones remain. Now, only when the new generation is

† We have ignored both thermal energies and the effect on the gravitational potential energy of the nucleosynthesis. This is excusable only as long as we are interested in an order of magnitude estimate of ϵ .

lighter than previous generations will there be new physics. Further, a new generation will only have a selective advantage if the quark masses are such as to allow for a spectrum of more than one stable bound state of 'nucleons'. However, this criterion is hard to meet. For example, it is not hard to work out what the physics would be were the second generation the lightest. The stable nucleon would be the Ω^- , and the gap between it and the next heaviest baryon (the neutral ssc) would be on the order of $m_{\text{charm}} - m_{\text{strange}} \approx 1 \text{ GeV}$, so that the Ω^{-1} would be the only stable nucleon. Such a universe would consist mostly of 'hydrogen' with a μ^+ bound to the Ω^- but there would be no nuclear fusion and, hence, if the previous scheme is right, no era of nucleosynthesis, no stars and no supernovas. It is not hard to see that the same situation would hold for the third generation.

The advantage of such a proposal is that it might explain both why there are many generations and why the lightest generation is so light. Because there is a narrow window in quark masses for which nuclear bound states exist, it must have taken many tries for an appropriate set to be produced randomly. Note, further, that such a proposal, while *ad hoc*, is easily falsifiable. An observation of a generation of quarks such that, were these the lightest generation, nuclear or other processes would exist which would increase the expected number of black holes, would falsify it.

8. Conclusions

There are many other aspects of particle physics that one could imagine trying to treat in this way. For example, is the smallness of the neutrino masses due in some part to the role that they play at the era of primordial nucleosynthesis [28] or in supernova? And finally and most importantly, there is the problem of the smallness of the cosmological constant.

In closing, I want to emphasize that the proposal made in this paper is extremely speculative. Its interest is that it asks us to do something new; rather than trying to tie all physics to some fundamental theory at the Planck scale the explanation for why a particular parameter takes a particular value is to be sought by asking how the rate that a universe produces black holes will be altered if that parameter is changed. However, it cannot be taken very seriously unless a detailed scenario and mechanism, based on known physics, can be developed to explain the value of a particular parameter which is falsifiable by some combination of experiment and theory. Whether this can be done successfully remains to be seen.

Acknowledgments

In the development of this proposal I have been influenced by three authors who made the problem of naturalness compelling to me in a way that suggested its affinity with the problem of fitness in biology: Greenstein [25], Lovelock [26] and Morowitz [27]. Yoichiro Nambu [28], in a paper I read several years ago, asks the question of whether there might be an evolutionary explanation for some problems in particle physics, including the problems of the generations and of the proton-neutron mass difference. Two lectures, one by Andrei Linde at GR12 [13] and one by George Sparling at the 1984 Oxford Quantum Gravity conference [29], had the effect of loosening my imagination about what was possible in cosmology. Finally, in working out these ideas

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